Feature: Phase 3 – Net Cash Flow Analysis Calculator

1. General Layout

- **Keep the current table layout** (no structural redesign)
- Only update labels, inputs, and calculation logic as described below

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2. Income Section

- Replace:
  - ❌ "Gross Annual Income"
  - ✅ With: **"Yearly Rent"**
- This represents total annual rental income

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3. Operating Expenses Section

- Show:
  - **"Operating Expenses"** as a single total value
- Add:
  - A clickable option:
    - **"Show cost breakdown"**

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4. Operating Expenses Breakdown (Expandable)

When user clicks "Show cost breakdown", display the following:

⚠️ Important:

- Users should see **only the calculated € values**, NOT percentages

Breakdown logic (based on yearly rent):

- Vacancy Costs → 4%
- Maintenance → 10%
- CapEx Reserve → 3%
- Insurance → 1.5%
- Property Tax → 8%
- Condominium Fees → 7%
- Management Fee → 10%
- Admin / Accounting → 1%

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5. Mortgage Section

Inputs:

- **Down Payment** (user input)
- **Interest Rate** (user input)

Auto-calculation:

- Calculate and display:
  - **LTV (Loan-to-Value)** based on:
    - Property price
    - Down payment

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6. Mortgage Disclaimer

- Display a note near the mortgage section:

  “This calculation assumes a full annuity mortgage.”

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7. Principal Toggle

- Add a toggle:
  - **“Include principal repayment in calculation”**

Behavior:

- If **OFF**:
  - Only include **interest payments**
- If **ON**:
  - Include:
    - Interest
    - Principal repayment

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8. Mortgage Cost Display

- Always show:
  - **Annual Interest** भुगतान
- Conditionally show (if toggle ON):
  - **Annual Principal Repayment**

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9. Profit Calculation

Output:

- **Annual Profit**
- **Monthly Profit**

Logic:

- Annual Profit =
  - Yearly Rent
  - − Operating Expenses
  - − Mortgage Costs (interest ± principal depending on toggle)
- Monthly Profit =
  - Annual Profit ÷ 12

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10. Layout Placement (Important)

Keep structure consistent with current UI:

1. Yearly Rent (top, replacing current income field)
2. Operating Expenses (with expandable breakdown)
3. Mortgage Costs:
   - Interest
   - (Optional) Principal
4. Final Outputs:
   - Annual Profit
   - Monthly Profit

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11. Notes / Implementation Details

- All percentages should be configurable for the users
- Breakdown values should update dynamically when rent changes
- Ensure smooth toggle behavior (no layout shift issues)
- Keep UX clean: breakdown hidden by default

Feature: Stage 4 – Capital Return (ROI)

1. Overall Goal

- **Present a clear view of capital deployment and return metrics**
- **Improve clarity of financial breakdown**

- **Add user control over including appreciation and principal repayment**

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2. Top Section – Capital Deployment (Left Card)

Keep the layout similar, but improve labeling and clarity:

Display the following with short explanatory text:

- **Total Project Cost (€X)**
  → Full cost including purchase price, taxes, and fees
- **Mortgage Loan (–€X)**
  → The financed portion (negative value)
- **Actual Cash Invested (€X)**
  → User's equity contribution

Additional Line:

- **Net Operating Income (NOI) (€X)**
  → Keep as is (already present)

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3. Top Section – Cash-on-Cash Return (Right Card)

- **Keep circular visualization**
- **Label:**
  - **Cash-on-Cash Return (%)**
  - **Subtitle:**
    "Annual return on invested cash (excl. appreciation by default)"

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4. Bottom Bar – Performance Metrics

A. Add Toggle

- **Add a toggle:**
  - **"Include appreciation & principal repayment"**

B. Metrics (Updated Logic)

1. Total Return on Equity

- **Dynamic based on toggle**

Formula:

- **If toggle OFF:**

  total_return = net_cash_flow / actual_cash_invested

- **If toggle ON:**

  total_return = (net_cash_flow + principal_repaid + appreciation) / actual_cash_invested

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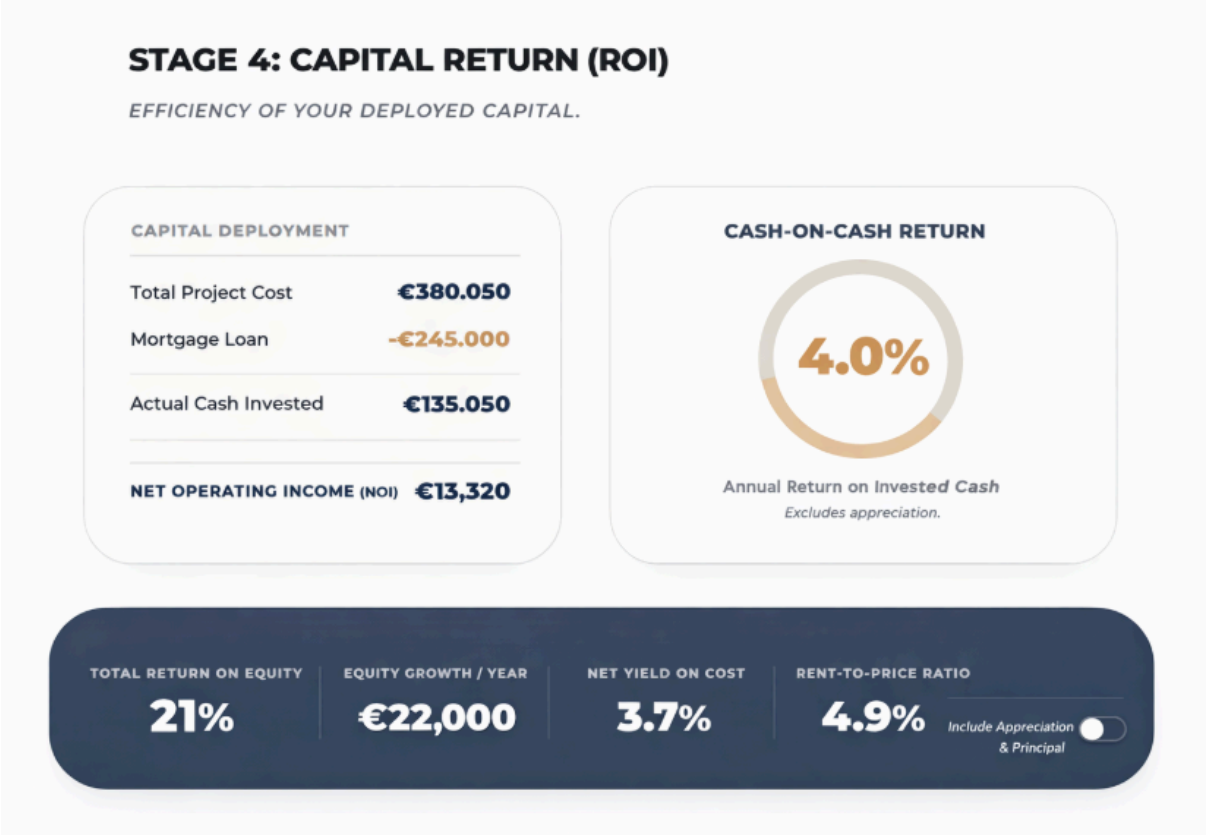

Use as reference but make a bit nicer and remove rent to price ratio

2. Equity Growth / Year

Formula:

equity_growth = principal_repaid + appreciation

- **If toggle OFF:**

    - Show only principal repaid OR show reduced value (depending on product decision)
- If toggle ON:
    - Show full value (principal + appreciation)

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3. Net Yield on Cost

- Keep unchanged
- No logic updates required

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4. Remove Metric

- ❌ Remove:
    - Rent-to-Price Ratio (completely)

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5. Visual / UX Improvements

- Make the design cleaner and more premium than current:
    - Better spacing between values
    - Stronger hierarchy (labels vs values)
    - Softer background contrast for cards
    - Ensure alignment between top cards and bottom bar
- Ensure:
    - Smooth toggle interaction (no layout jump)
    - Values update instantly when toggled

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6. Data Dependencies

This stage depends on:

- Net Cash Flow (from Stage 3)
- Principal Repaid (mortgage calculation)
- Appreciation (user input or predefined assumption)
- Actual Cash Invested (Stage 4 calculation)

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7. Notes

- Keep all formulas modular (for future ROI extensions like IRR)

- Ensure currency formatting is consistent (e.g. €380,050)
- Percentages should be rounded appropriately (e.g. 4.0%)

1. Biggest Issue: “Growth Weighting” Is Too Abstract

The “Growth weighting” slider (50%) is not something serious investors think in.

- It’s unclear what it actually does mathematically
- It mixes two fundamentally different return drivers:
  - Appreciation (market-driven)
  - Rental income (operational)

✅ Replace it with something concrete:

Option A (recommended):

- ❌ Remove “Growth weighting”
- ✅ Replace with:
  - “Annual rent growth (%)”

Now you have:

- Yearly appreciation (%)
- Yearly rent growth (%)

This is much more realistic and intuitive.

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2. Appreciation Input Is Good — But Needs Context

You currently have:

- “Yearly appreciation (%) – 6%”

That’s fine, but:

Improve it:

- Add a range hint or benchmark
  - e.g. “Typical range: 3% – 6% depending on market”
- Add tooltip:
  - “Based on historical property price growth in this market”

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3. The Chart Needs Meaning, Not Just Aesthetic

Right now the bar chart looks nice but doesn't clearly communicate:

- **What's growing?**
- **What components are included?**

Improve:

Split the chart into stacked growth:

- **Rental income (cumulative)**
- **Property value appreciation**

Or:

- **Add a toggle:**
  - **"Show breakdown"**

This makes it feel like a real investment tool, not just a visual.

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4. Connect This Phase to Earlier Phases

Right now Phase 5 feels slightly disconnected.

Fix:

Make sure inputs are clearly tied to:

- **Net cash flow (from Phase 3)**
- **ROI logic (from Phase 4)**

Add a small label:

> **"Projection based on your current investment setup"**

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5. ROI Number (407%) Needs Transparency

"Combined 20Y ROI – 407%" is powerful but risky if unclear.

Add breakdown:

Under it, show:

- **Rental income contribution: €X**
- **Appreciation contribution: €X**

Or at least:

- **% from rent**
- **% from appreciation**

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6. Add Time Horizon Control (High Impact)

You hardcoded:

- **“Project your investment over 20 years”**

Better:

- **Add selector:**
  - **10 / 15 / 20 / 25 / 30 years**

This dramatically improves usefulness.

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7. Include Financing Reality (Important)

Right now this phase ignores financing dynamics.

Add (optional toggle):

- **“Include mortgage payoff over time”**
  - **This affects:**
    - **Equity growth**
    - **Total return**

This ties perfectly with Stage 4.

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8. Improve the “Valuation Logic” Box

Current text is a bit generic.

Make it sharper:

Instead of:

> **“Appreciation is less predictable…”**

Use something more actionable:

> "Returns come from two sources:
> rental income (stable) and appreciation (market-driven).
> Adjust assumptions to reflect your risk expectations."

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9. Add One Missing Metric (Very Important)

You're missing:

✅ IRR (Internal Rate of Return)

This is the metric serious investors expect in long-term projections.

Add:

- "Projected IRR: X%"

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10. Small UX Improvements

- Make sliders show exact value while dragging
- Add € formatting everywhere (consistent with earlier stages)
- Animate chart when values change (subtle, not flashy)

What I Would Remove

- ❌ Growth weighting (too abstract)
- ❌ Any "magic blending" between rent & appreciation

What I Would Add (Summary)

- ✅ Rent growth %
- ✅ Time horizon selector
- ✅ ROI breakdown (rent vs appreciation)
- ✅ IRR metric
- ✅ Optional mortgage payoff inclusion
- ✅ Clearer chart meaning